Green University of Bangladesh
Department of Computer Science and Engineering (CSE)

Faculty of Sciences and Engineering
Semester: (Spring, Year: 2026), B.Sc. in CSE (Day)

Lab Report NO: 01
Course Title: AI Lab
Course Code: CSE 316
Section: 232-D5

Lab Experiment Name: Introduction to Basic Operations on Python

Student Details

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Lab Date : **08/02/2026**
Submission Date : **14/02/2026**
Course Teacher's Name : **Babe Sultana**

Lab Report Status

Marks: …… **Signature:.....**
Comments:..... **Date:.....**

1. INTRODUCTION

In this lab, we practice solving different problems using Python. The tasks include mathematical operations, list processing, and function implementation. This helps build a strong foundation in programming logic and basic coding skills.

2. OBJECTIVES/AIM

- To learn basic operations in python
- To developed logic in python

3. PROCEDURE

In this lab, multiple programming problems were solved using Python by applying fundamental concepts and logical techniques. The separation of odd and even numbers was performed using the modulus operator, while the smallest value in a list was identified through continuous comparison. A specific range of numbers was filtered based on divisibility conditions to compute a selective sum, and the second-largest number was obtained by carefully updating two tracking variables. The factorial was determined through repeated multiplication, and the Fibonacci sequence was produced by successively adding the two preceding terms.

Functions were used to compare two numbers and return the larger one, as well as to calculate totals from given parameters. List operations included computing the total sum, identifying the maximum value, counting positive, negative, and zero elements, and calculating the average. Prime numbers were validated through divisibility testing within a defined range. Common multiples of 4 and 6 were displayed based on their shared divisibility condition. Additional tasks involved determining the greatest among three numbers, checking whether a number is even or odd, eliminating duplicate entries from a list, and finding the difference between maximum and minimum values. Finally, dictionary operations were carried out to locate the key associated with the highest value and to merge two dictionaries into a single combined structure.

4. IMPLEMENTATION:

Code:

Problem1:

```
numbers = {1, 2, 3, 4, 5, 6, 7, 8, 9}

even_sum = 0
odd_sum = 0

for value in numbers:
    if value % 2 == 0:
        even_sum += value
```

```
    else:
        odd_sum += value

print("Even sum =", even_sum)
print("Odd sum =", odd_sum)
```

Problem2:

```
nums = [8, 3, 12, 5, 2]

smallest = nums[0]

for value in nums[1:]:
    if value < smallest:
        smallest = value

print("Smallest number =", smallest)
```

Problem3:

```
total = 0

for num in range(50, 101):
    if num % 3 == 0:
        if num % 5 != 0:
            total += num

print("Total sum =", total)
```

Problem4:

```
arr = [10, 4, 8, 15, 6]

first = second = -999999

for value in arr:
    if value > first:
        second = first
        first = value
    elif value > second and value != first:
        second = value

print("Second highest =", second)
```

Problem5:

```
number = int(input("Enter number: "))
result = 1

for i in range(number, 0, -1):
    result *= i

print("Factorial =", result)
```

Problem6:

```
terms = int(input("Enter number of terms: "))

x, y = 0, 1

for _ in range(terms):
    print(x, end=" ")
    x, y = y, x + y
```

Problem7:

```
def get_largest(x, y):
    return x if x > y else y

a = int(input("Enter first: "))
b = int(input("Enter second: "))

print("Largest =", get_largest(a, b))
```

Problem8:

```
def add_numbers(*values):
    total = 0
    for v in values:
        total += v
    return total

print("Sum =", add_numbers(5, 10, 15, 20))
```

Problem9:

```python
items = [2, 4, 6, 8]

total = 0
index = 0

while index < len(items):
    total += items[index]
    index += 1

print("List sum =", total)
```

Problem10:

```python
values = [11, 25, 7, 19]

largest = values[0]

for i in range(1, len(values)):
    if values[i] > largest:
        largest = values[i]

print("Largest =", largest)
```

Problem11:

```python
data = [5, -3, 0, 8, -1, 0]

pos = neg = zero = 0

for value in data:
    if value > 0:
        pos += 1
    elif value < 0:
        neg += 1
    else:
        zero += 1

print("Positive:", pos)
print("Negative:", neg)
print("Zero:", zero)
```

Problem12:

```python
n = int(input("Enter number: "))

if n <= 1:
    print("Not Prime")
else:
    is_prime = True
    for i in range(2, n // 2 + 1):
        if n % i == 0:
            is_prime = False
            break

    print("Prime" if is_prime else "Not Prime")
```

Problem13:

```python
for num in range(1, 101):
    if num % 12 == 0:
        print(num)
```

Problem14:

```python
def max_of_three(a, b, c):
    if a >= b and a >= c:
        return a
    elif b >= a and b >= c:
        return b
    else:
        return c

print("Maximum =", max_of_three(9, 14, 11))
```

Problem15:

```python
def is_even(n):
    return n % 2 == 0

number = 12

if is_even(number):
    print("Even")
else:
    print("Odd")
```

Problem16:

```
numbers = [1, 2, 2, 3, 4, 3, 5]

new_list = []

for value in numbers:
    if value not in new_list:
        new_list += [value]

print("Without duplicates:", new_list)
```

Problem17:

```
nums = [3, 6, 9, 12]

total = sum(nums)
count = len(nums)

avg = total / count

print("Average =", avg)
```

Problem18:

```
values = [20, 5, 15, 30]

maximum = max(values)
minimum = min(values)

print("Difference =", maximum - minimum)
```

Problem19:

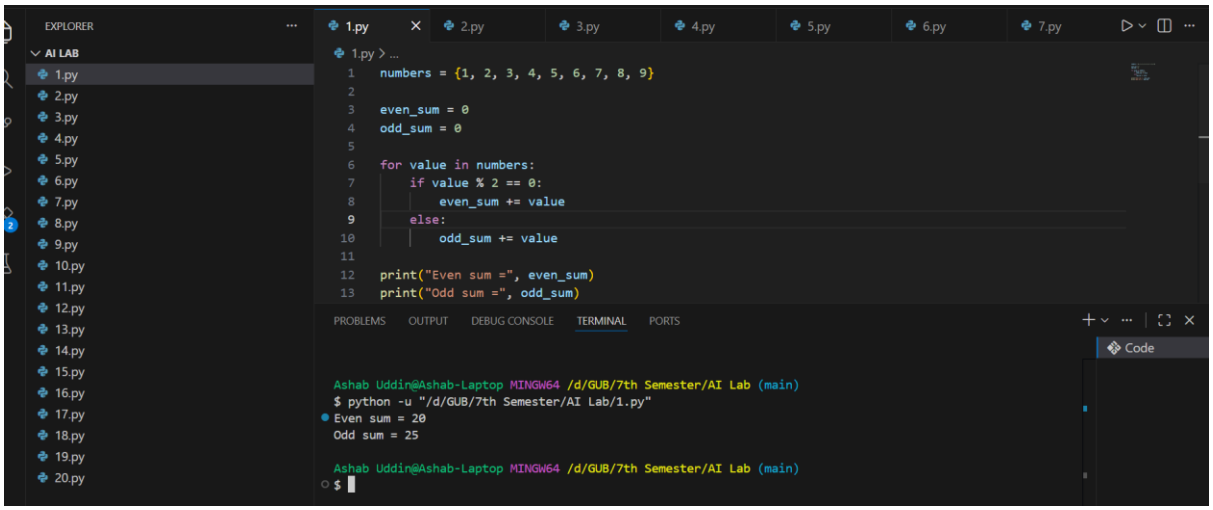
```
info = {"x": 50, "y": 75, "z": 60}

highest_key = None
highest_value = -1

for key, value in info.items():
    if value > highest_value:
        highest_value = value
        highest_key = key

print("Key with max value:", highest_key)
```

Problem20:

```
d1 = {"a": 100, "b": 200}
d2 = {"c": 300, "d": 400}

combined = d1.copy()

for key, value in d2.items():
    combined[key] = value

print("Merged dictionary:", combined)
```

5. OUTPUTS:

Problem1:



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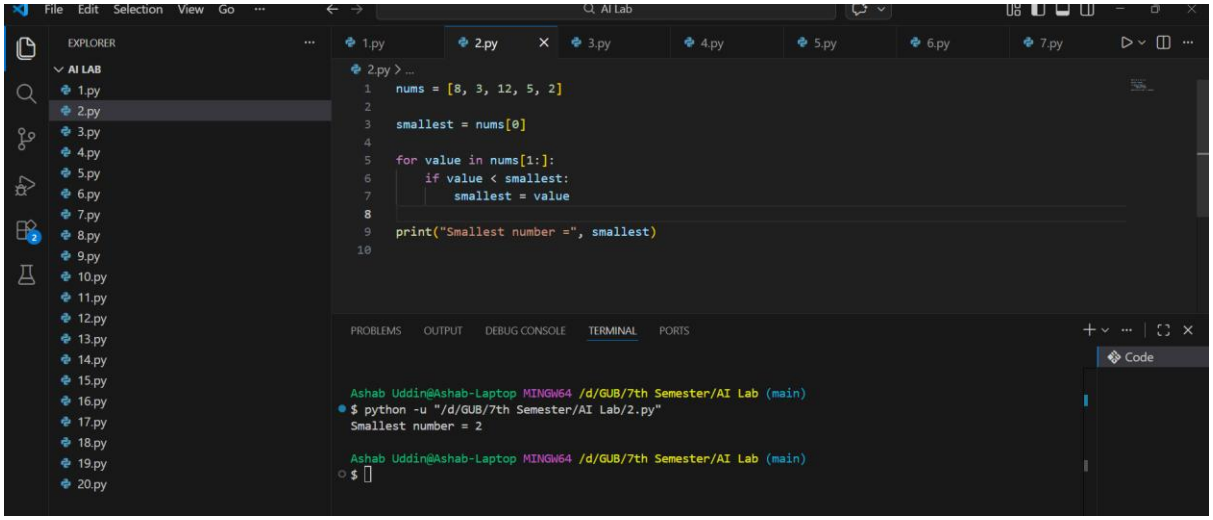
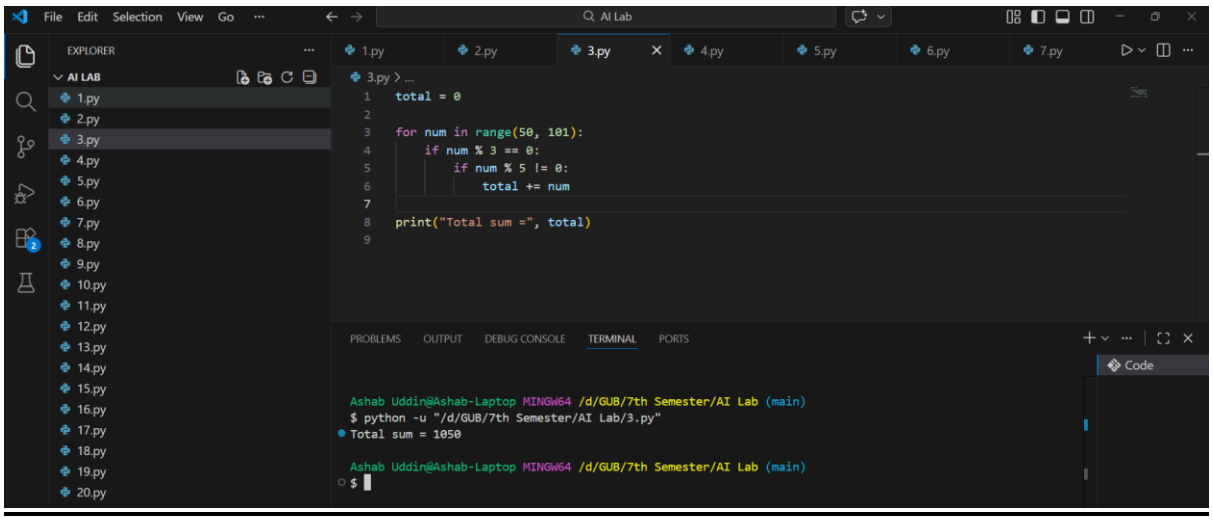


Problem3:



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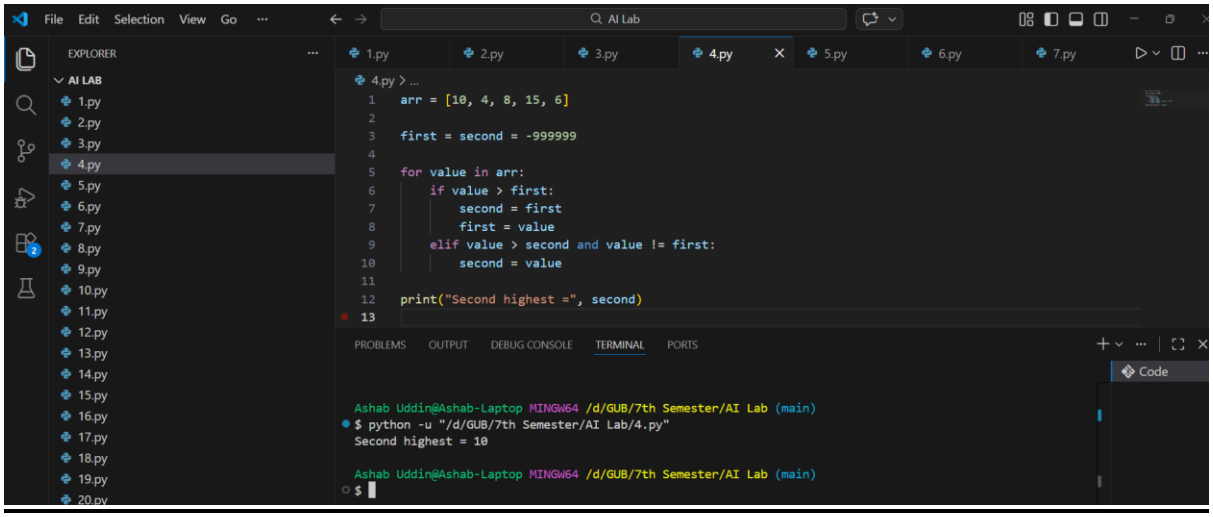

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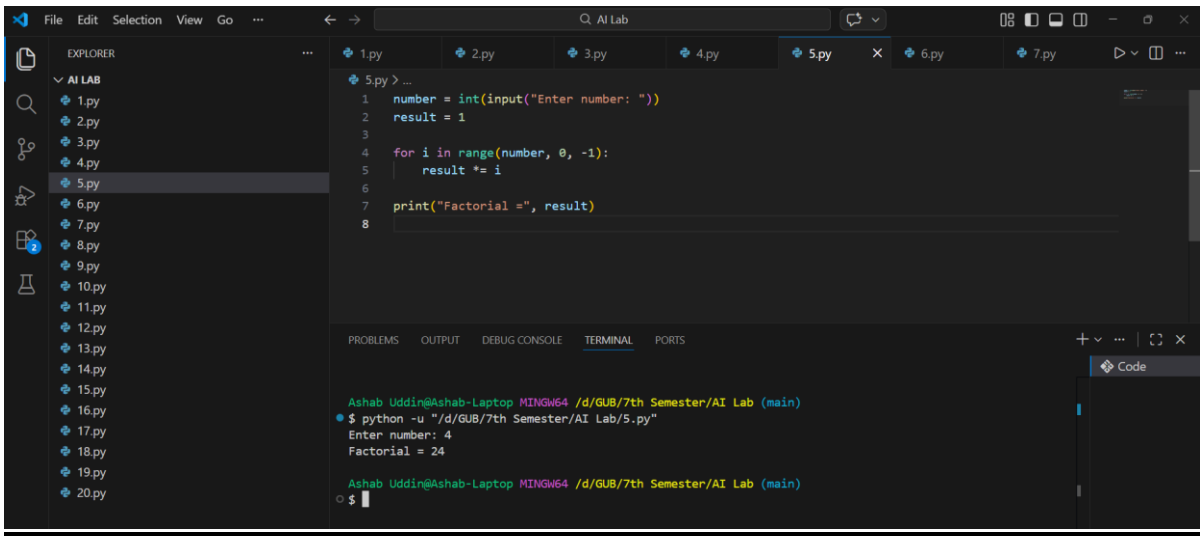

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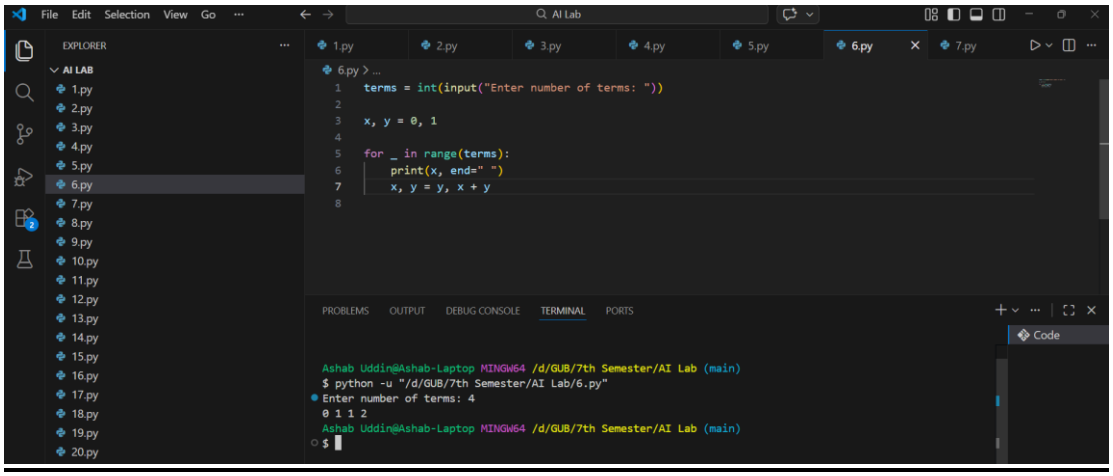

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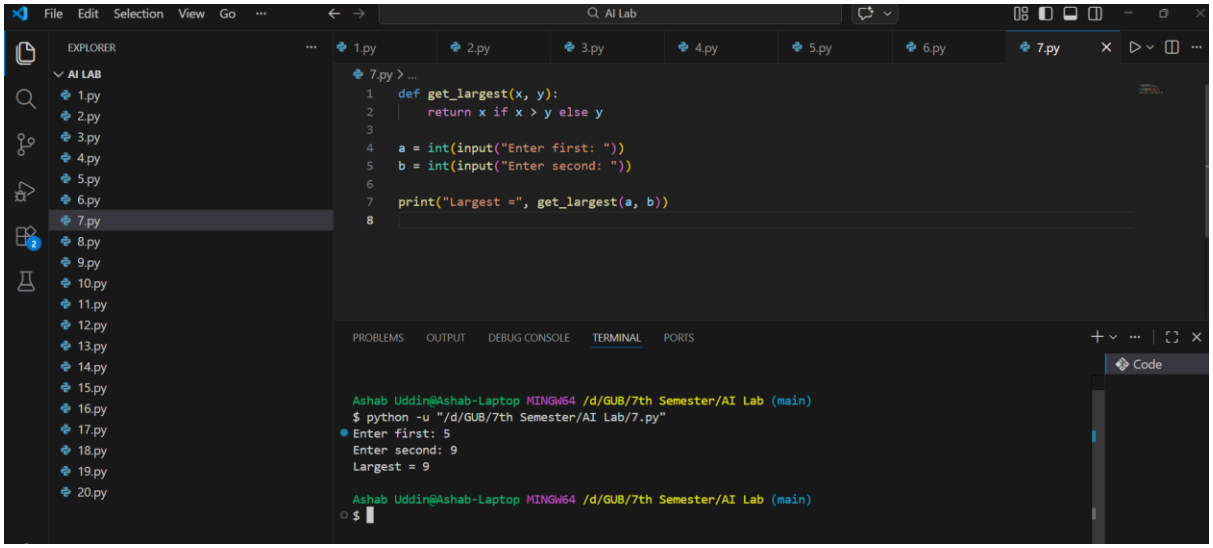

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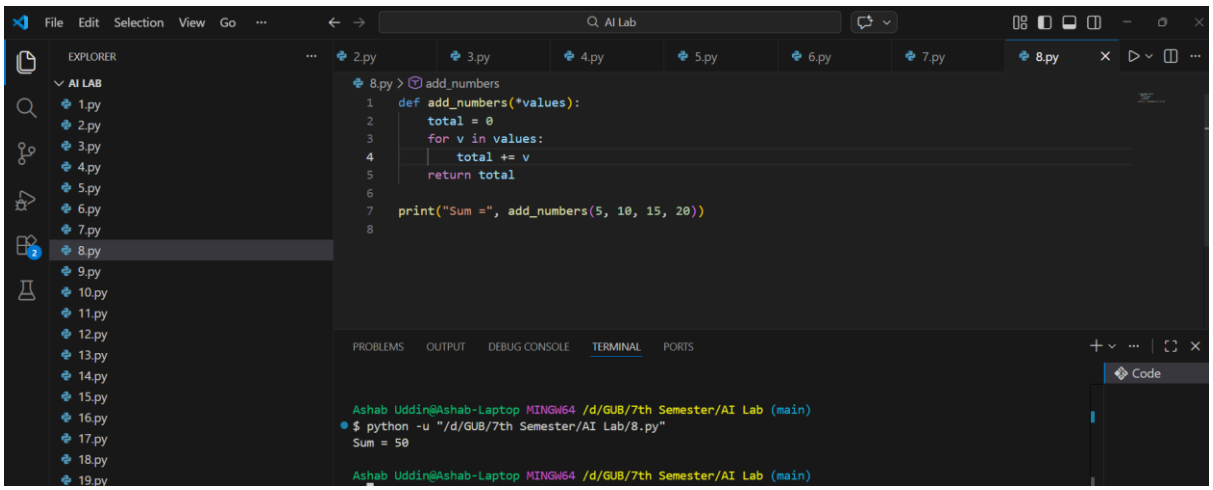

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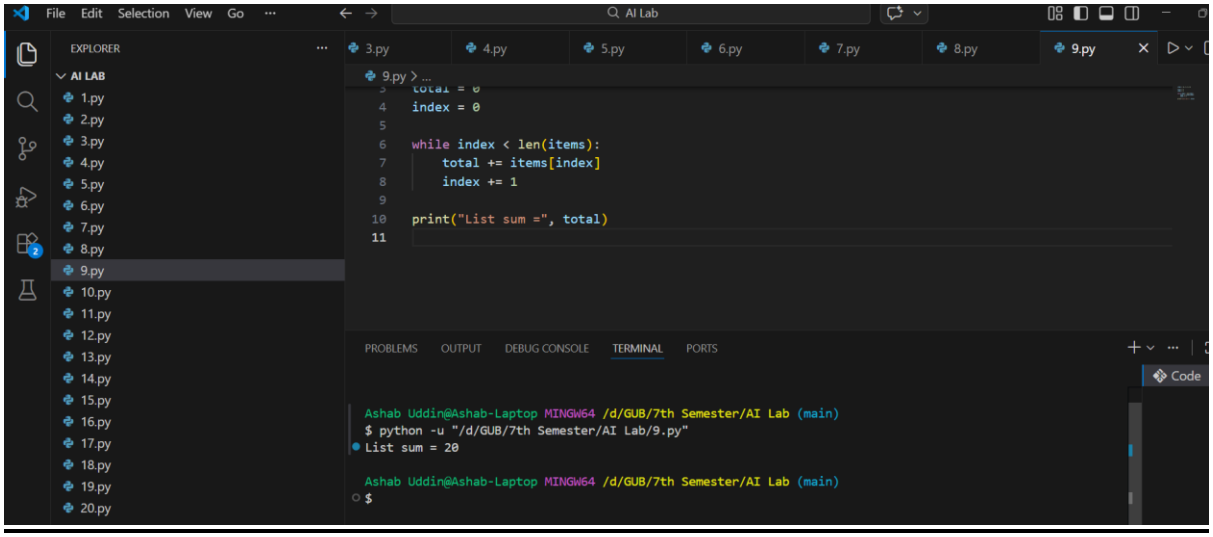

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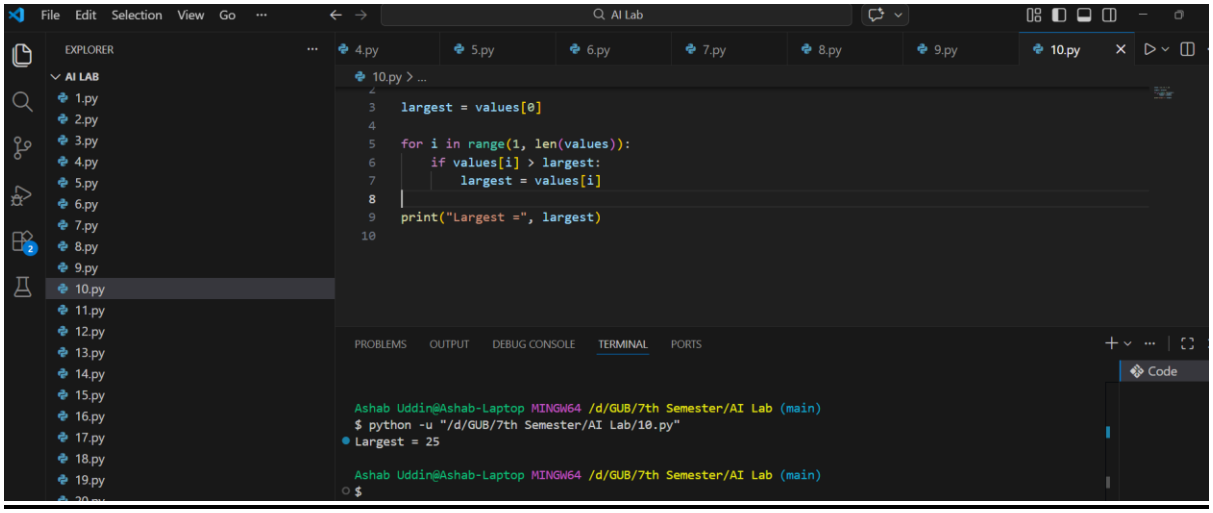

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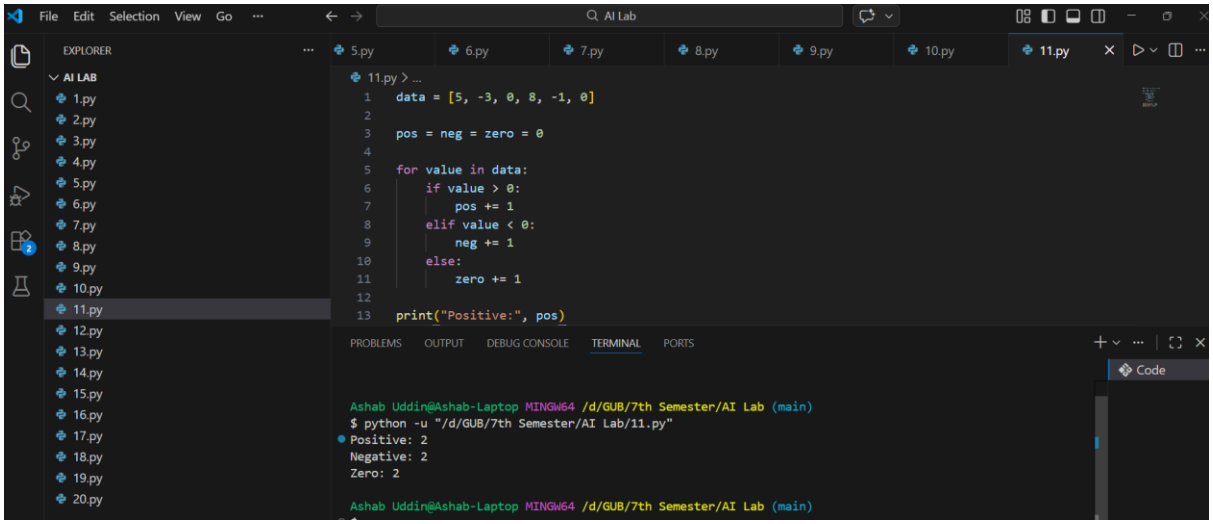

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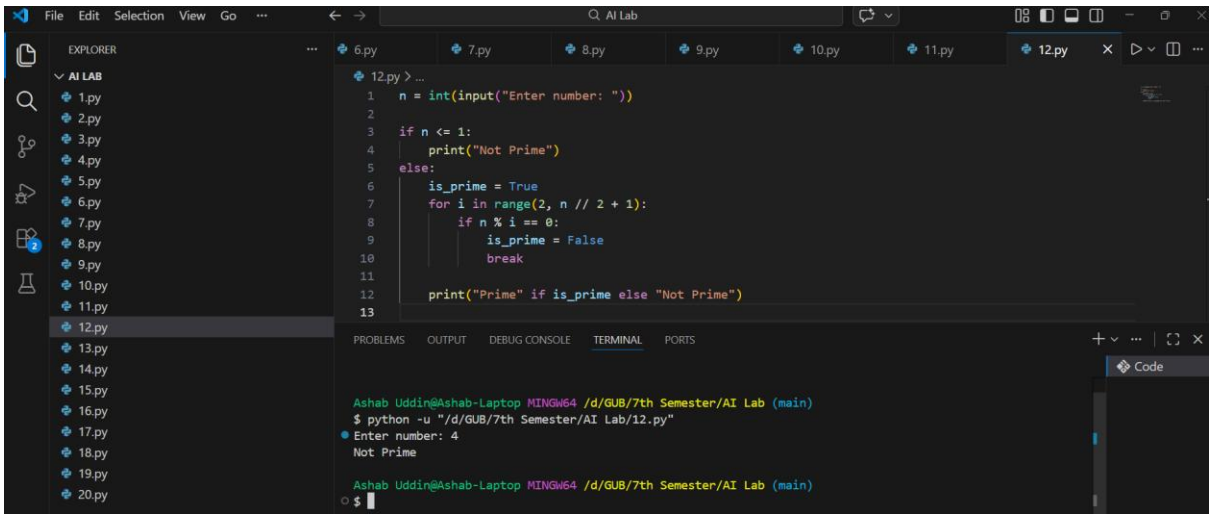

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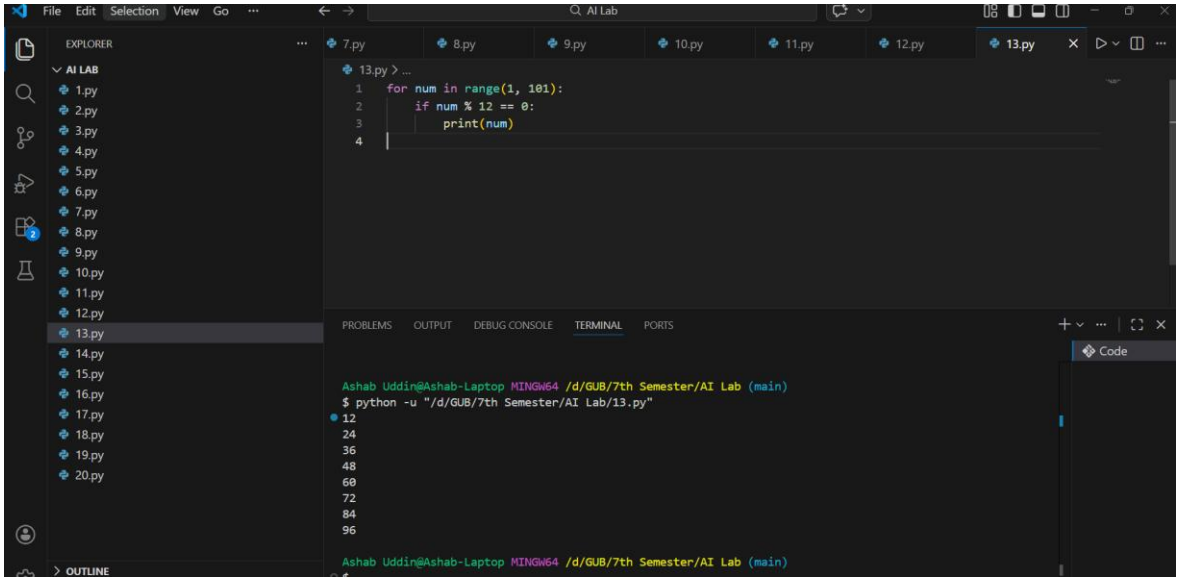

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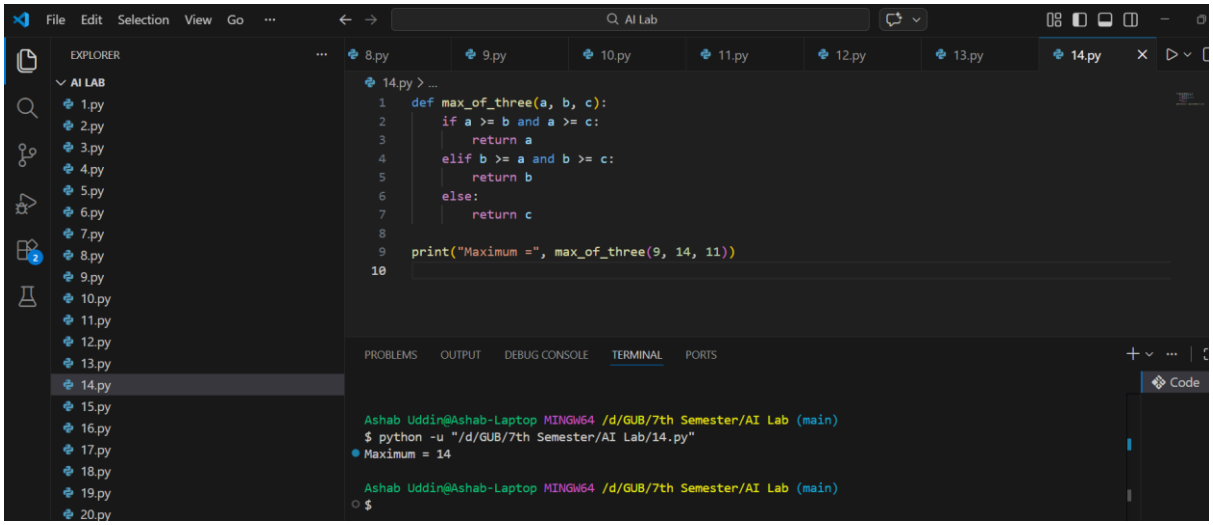

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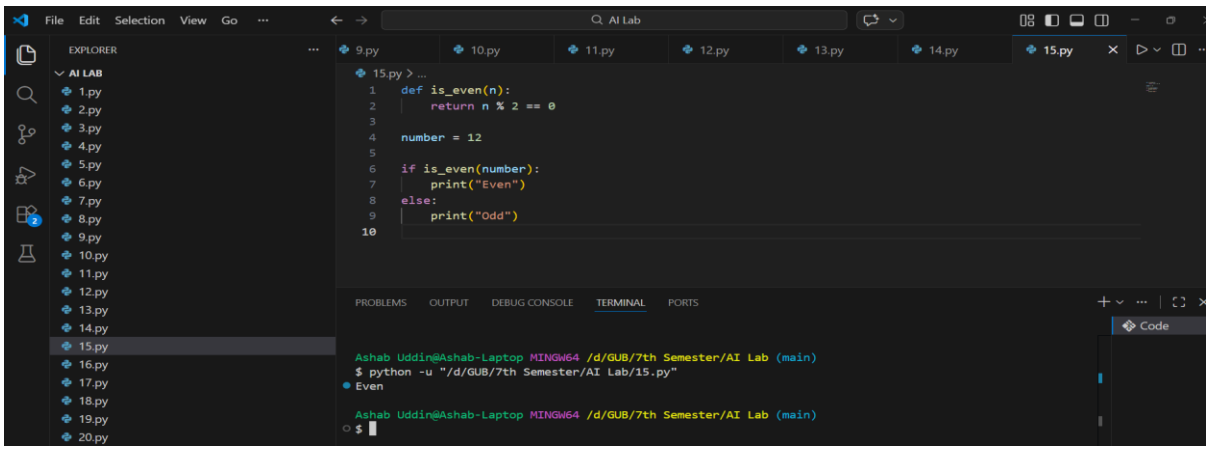

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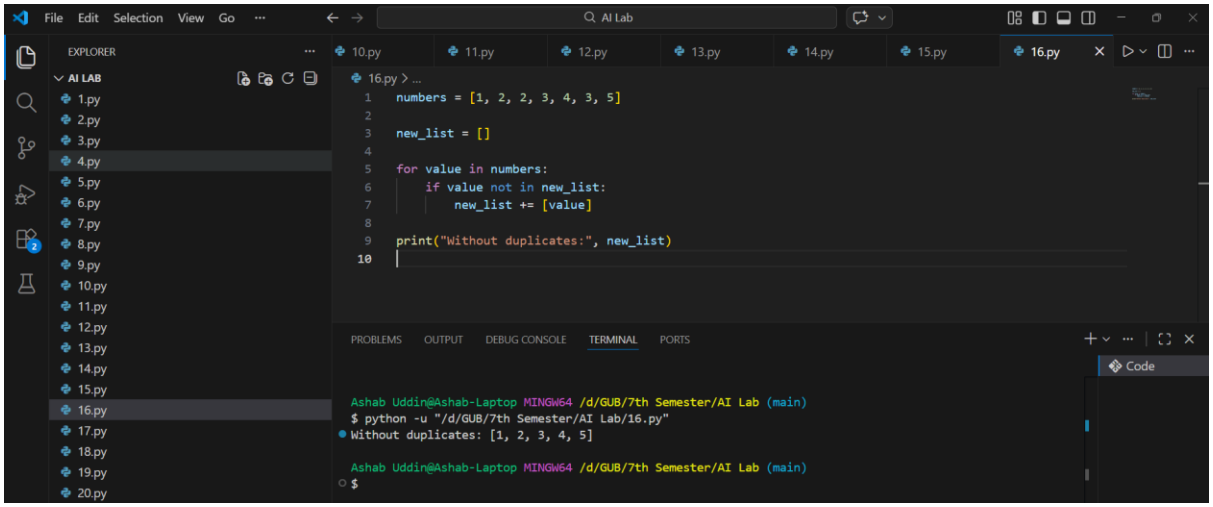
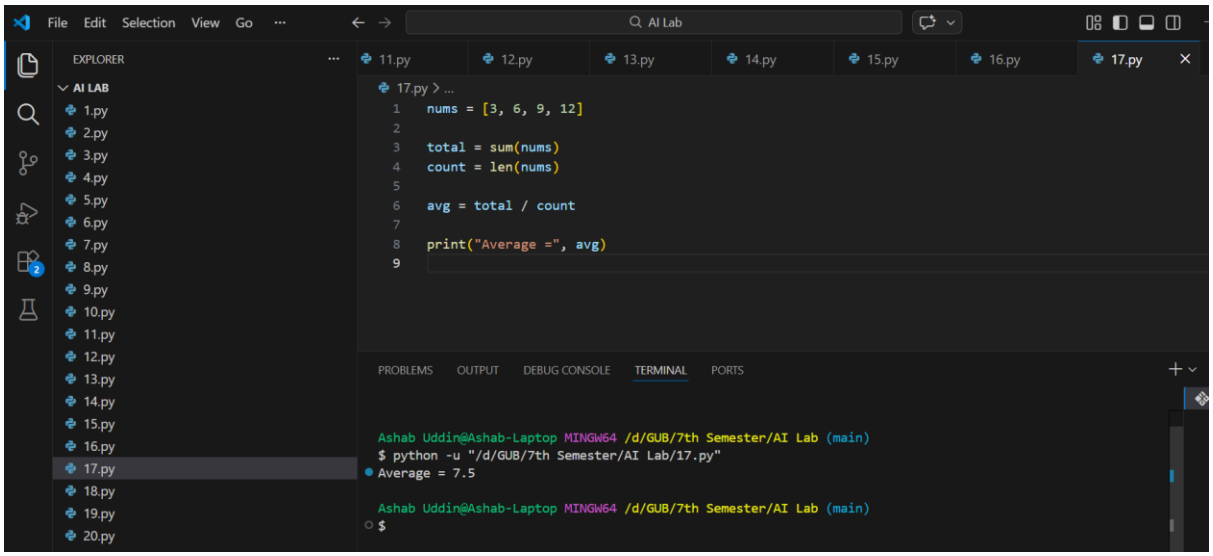


Problem17:



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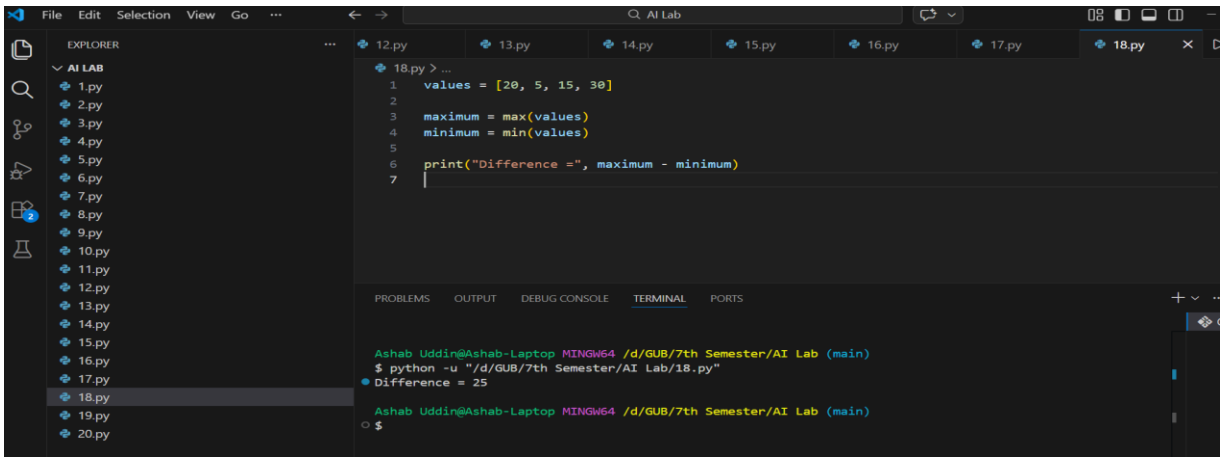

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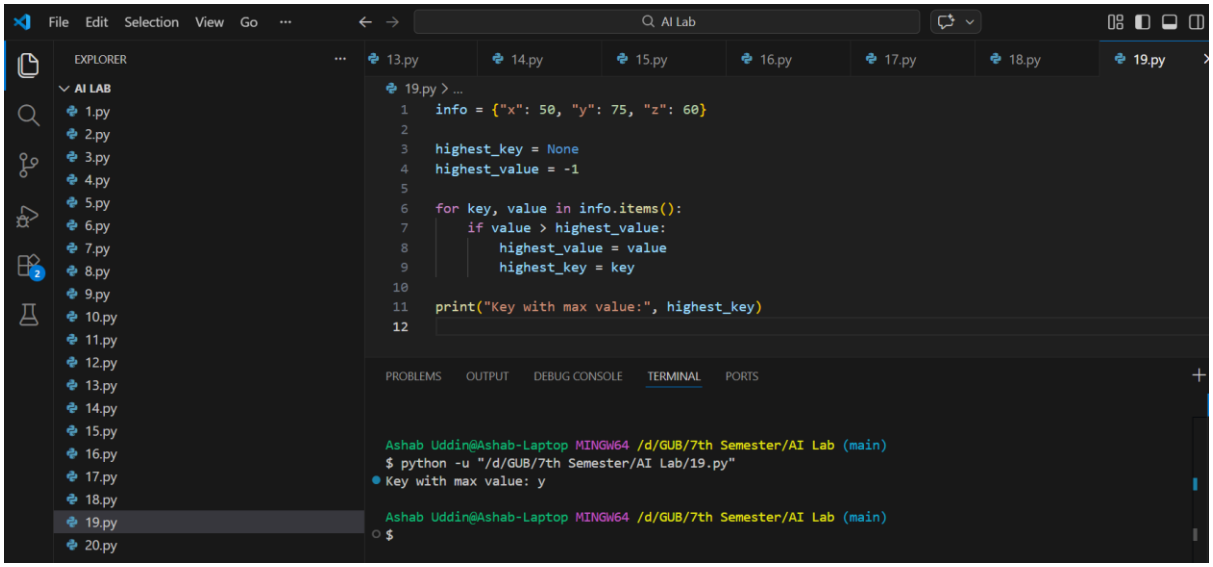
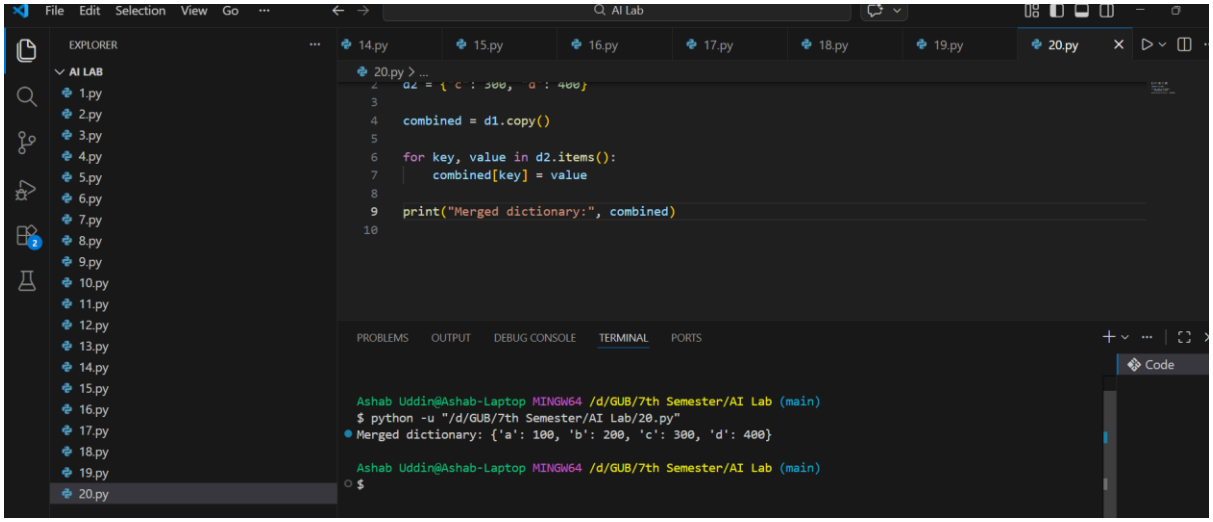


Problem20:



6. ANALYSIS AND DISCUSSION:

In this lab, we applied basic Python concepts like loops, conditionals, functions, lists, sets, and dictionaries to solve different problems. Tasks like calculating sums, factorials, Fibonacci series, and finding maximum or minimum values helped strengthen logical thinking and iterative problem-solving.

Using functions made the code organized and reusable, while handling lists and dictionaries showed how to store, access, and manipulate data efficiently. Overall, these exercises provided a solid foundation in programming and prepared us for more advanced coding challenges.